

Support Vector Machine (SVM) for Diabetes Prediction

Dataset: Pima Indians Diabetes

Presentated by Owen & Sam

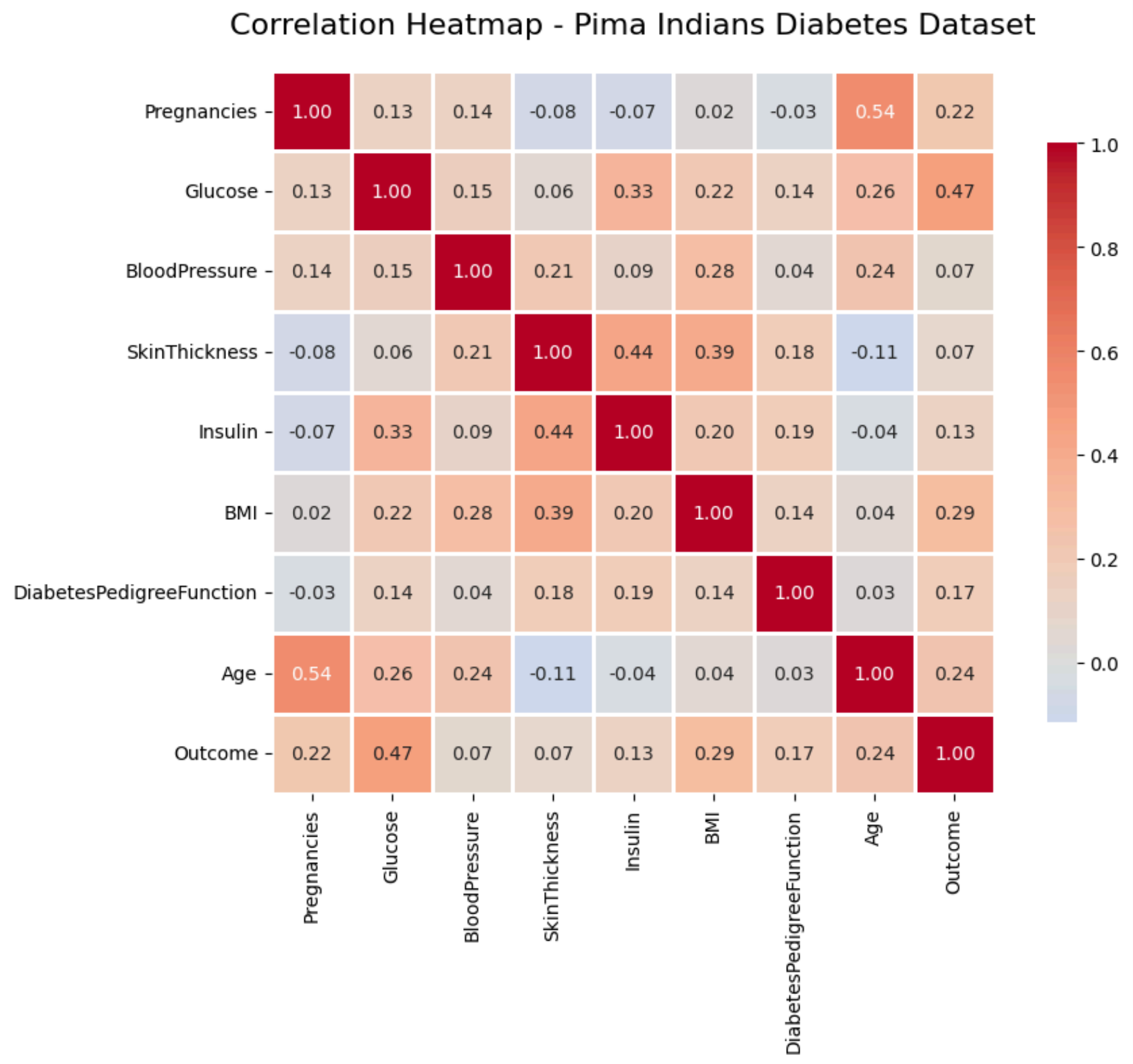
Introduction

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1
5	5	116	74	0	0	25.6	0.201	30	0
6	3	78	50	32	88	31.0	0.248	26	1
7	10	115	0	0	0	35.3	0.134	29	0
8	2	197	70	45	543	30.5	0.158	53	1
9	8	125	96	0	0	0.0	0.232	54	1

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Pregnancies            768 non-null    int64  
1   Glucose                768 non-null    int64  
2   BloodPressure          768 non-null    int64  
3   SkinThickness          768 non-null    int64  
4   Insulin                768 non-null    int64  
5   BMI                    768 non-null    float64 
6   DiabetesPedigreeFunction 768 non-null    float64 
7   Age                    768 non-null    int64  
8   Outcome                768 non-null    int64  
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
```



Introduction



Looking for error in the dataset

count	# Glucose	# BloodPressure	# SkinThickness	# Insulin	# BMI
mean	768.0	768.0	768.0	768.0	768.0
std	120.89453125	69.10546875	20.536458333333332	79.79947916666667	31.992578124999998
min	31.97261819513622	19.355807170644777	15.952217567727637	115.24400235133817	7.884160320375446
25%	0.0	0.0	0.0	0.0	0.0
50%	99.0	62.0	0.0	0.0	27.3
75%	117.0	72.0	23.0	30.5	32.0
max	140.25	80.0	32.0	127.25	36.6
	199.0	122.0	99.0	846.0	67.1

Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
DiabetesPedi	0
Age	0
Outcome	0

The redflag error dataset

```
Zero counts in each column:
```

Pregnancies	111
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11
DiabetesPedigreeFunction	0
Age	0
Outcome	500

```
dtype: int64
```

Handling the 0 value

Age $\geq$ 40

	Glucose	BloodPressure
count	207.000000	207.000000
mean	131.053140	75.845411
std	33.306055	17.046346
min	0.000000	0.000000
25%	106.000000	70.000000
50%	129.000000	78.000000
75%	154.500000	84.000000
max	197.000000	114.000000

Age $<$ 40

	Glucose	BloodPressure
count	561.000000	561.000000
mean	117.146168	66.618538
std	30.656682	19.576880
min	0.000000	0.000000
25%	96.000000	60.000000
50%	112.000000	70.000000
75%	133.000000	78.000000
max	199.000000	122.000000

Imputation of BloodPressure & Glucose

```
=== Imputation Values ===  
Age >= 40:  
  Glucose median: 129.00  
  BloodPressure mean: 78.11  
  
Age < 40:  
  Glucose median: 113.00  
  BloodPressure mean: 70.25  
  
=== Number of Zeros Filled ===  
Glucose zeros before: 5  
Glucose zeros after: 0  
BloodPressure zeros before: 35  
BloodPressure zeros after: 0
```

Imputation of BMI

# Glucose	# BloodPressure	# SkinThickness	# Insulin	# BMI
768.0	768.0	768.0	768.0	768.0
120.89453125	69.10546875	20.536458333333332	79.79947916666667	31.992578124999998
31.97261819513622	19.355807170644777	15.952217567727637	115.24400235133817	7.884160320375446
0.0	0.0	0.0	0.0	0.0
99.0	62.0	0.0	0.0	27.3
117.0	72.0	23.0	30.5	32.0
140.25	80.0	32.0	127.25	36.6
199.0	122.0	99.0	846.0	67.1

BMI	
count	768.000000
mean	32.455208
std	6.875177
min	18.200000
25%	27.500000
50%	32.300000
75%	36.600000
max	67.100000

Skin Thickness

“Skin thickening is frequently observed in patients with diabetes. Affected areas of skin can appear thickened, waxy, or edematous. These patients are often asymptomatic but can have a reduction in sensation and pain. Although different parts of the body can be involved, the hands and feet are most frequently involved. Ultrasound evaluation of the skin can be diagnostic and exhibit thickened skin. Subclinical generalized skin thickening is the most common type of skin thickening. Diabetic thick skin may represent another manifestation of scleroderma-like skin changes, limited joint mobility, or scleredema diabeticorum, which are each described in more detail below.”

Edwards, Elise, and Gil Yosipovitch. "Skin Manifestations of Diabetes Mellitus." Endotext [Internet]. MDText. com, Inc., 2025.

Imputation of SkinThickness



Diabetes

	SkinThickness
count	268.000000
mean	22.164179
std	17.679711
min	0.000000
25%	0.000000
50%	27.000000
75%	36.000000
max	99.000000

No Diabetes

	SkinThickness
count	500.000000
mean	19.664000
std	14.889947
min	0.000000
25%	0.000000
50%	21.000000
75%	31.000000
max	60.000000

Imputation of SkinThickness



```
=== Imputation Values ===  
Mean SkinThickness for Outcome 0 (No Diabetes): 27.24  
Mean SkinThickness for Outcome 1 (Diabetes): 33.00  
  
=== Zeros Filled ===  
SkinThickness zeros before: 227  
SkinThickness zeros after: 0
```

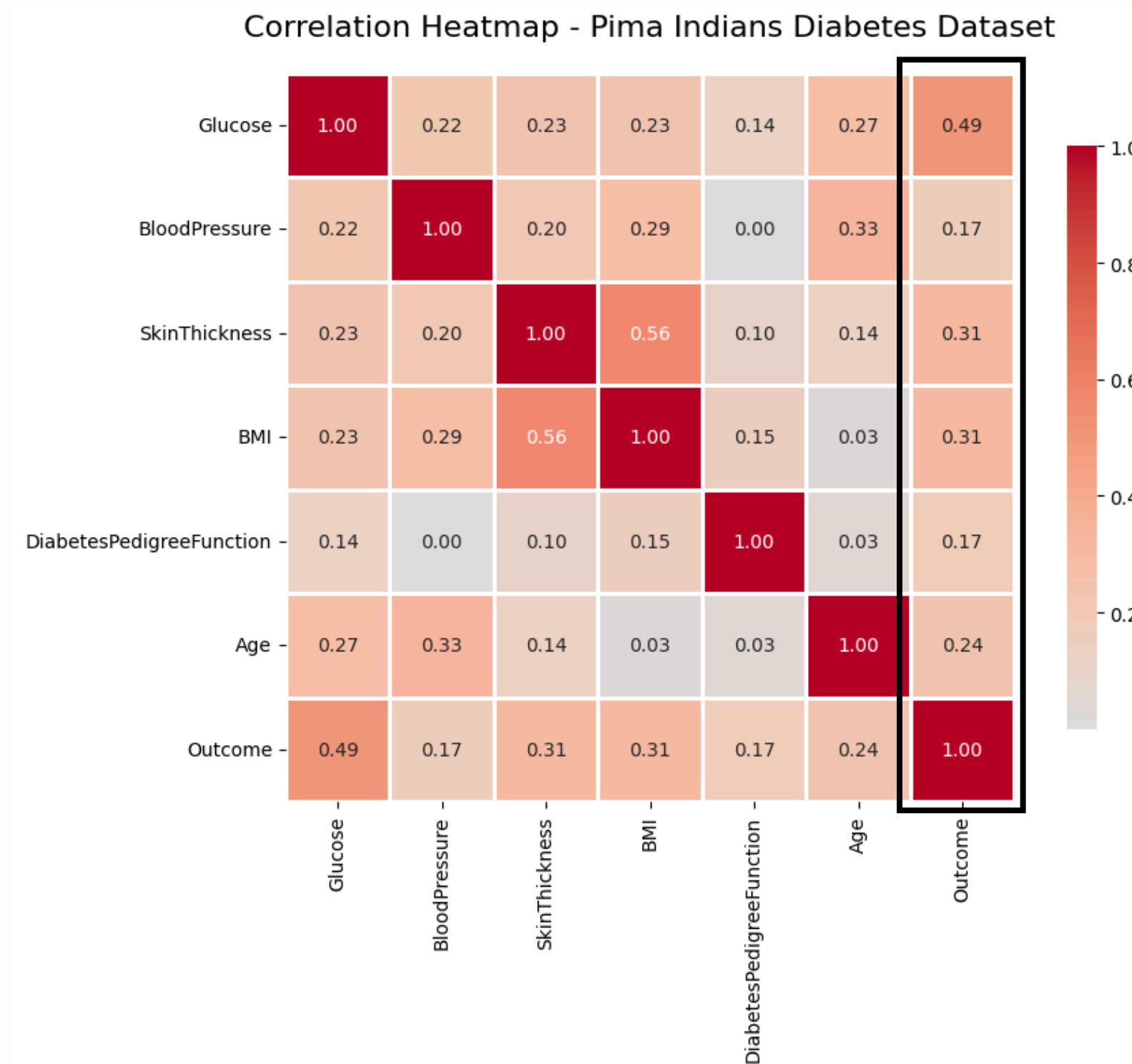
Cleaned dataset part 1

	# Pregnancies	# Glucose	# BloodPressure	# SkinThickness	# Insulin	# BMI	# DiabetesPedigreeFunction	# Age	# Outcome
count	268.0	268.0	268.0	268.0	268.0	268.0	268.0	268.0	268.0
mean	4.865671641791045	142.16044776119404	75.1066356649588	22.16417910447761	100.33582089552239	35.383582089552235	0.5505	37.06716417910448	1.0
std	3.7412390440415546	29.5534873919682	11.979738061939667	17.679711400465692	138.68912473153497	6.595602825647213	0.3723544835546108	10.968253652367915	0.0
min	0.0	78.0	30.0	0.0	0.0	22.9	0.088	21.0	1.0
25%	1.75	119.0	68.0	0.0	0.0	30.9	0.2625	28.0	1.0
50%	4.0	140.0	74.0	27.0	0.0	34.25	0.449	36.0	1.0
75%	8.0	167.0	82.0	36.0	167.25	38.775000000000006	0.728	44.0	1.0
max	17.0	199.0	114.0	99.0	846.0	67.1	2.42	70.0	1.0

```
Pregnancies      111
Glucose           0
BloodPressure     0
SkinThickness     0
Insulin          374
BMI              0
DiabetesPedigreeFunction  0
Age              0
Outcome          500
dtype: int64
```

Dropping unused 'label'

Columns remaining in features: ['Glucose', 'BloodPressure', 'SkinThickness', 'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome']

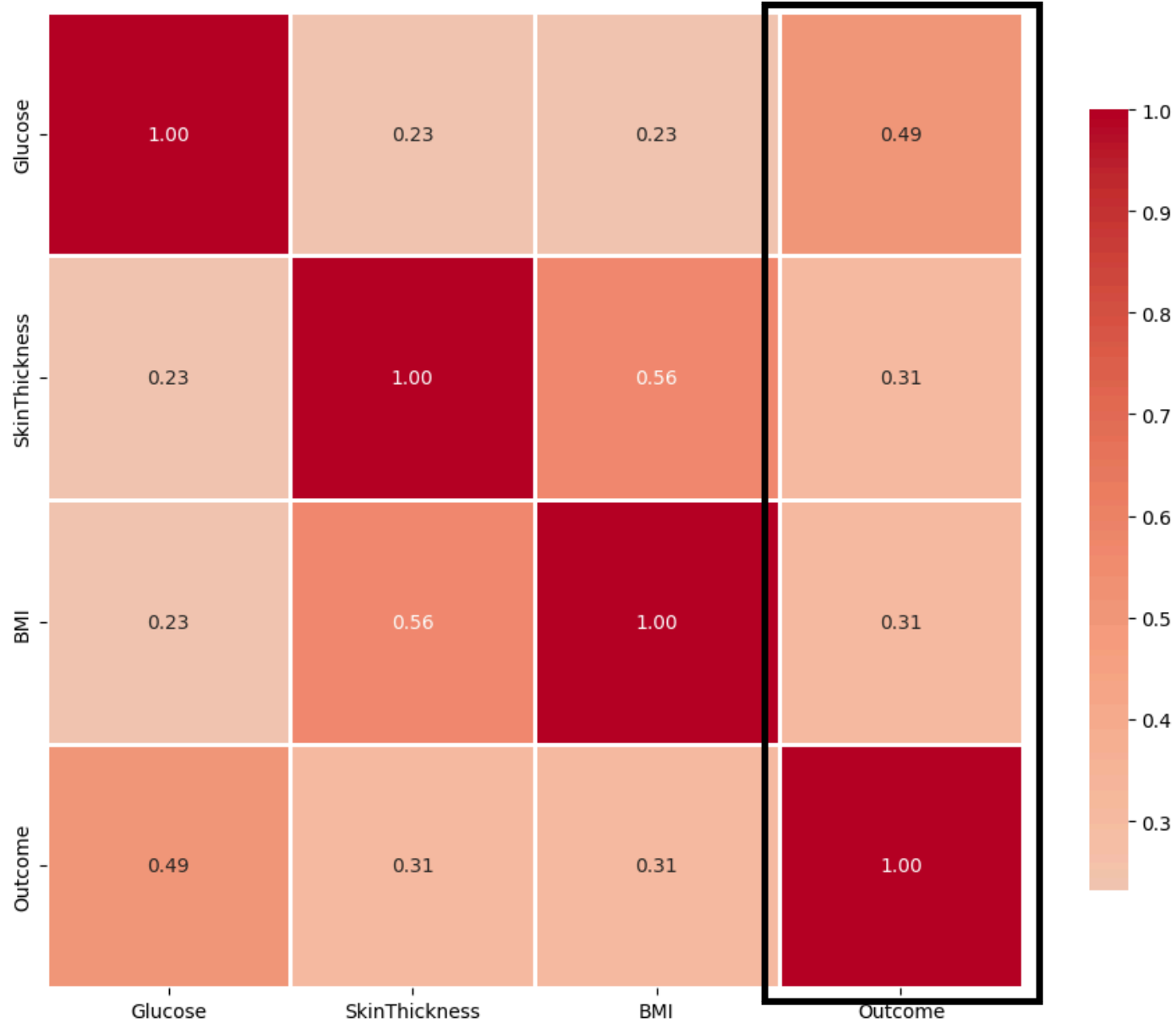


Dropping possible redflag for SVM 'label'

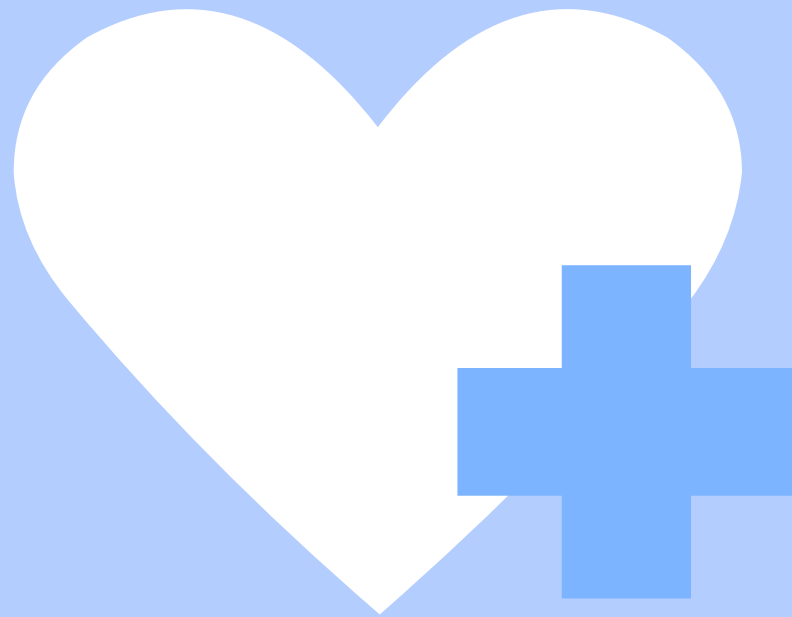
Columns remaining in features: ['Glucose', 'SkinThickness', 'BMI', 'Outcome']

	# Glucose	# SkinThickness	# BMI	# Outcome
590	111	40.0	46.8	1
200	113	16.0	31.0	0
507	130	23.0	28.6	0
139	105	29.0	36.9	0
351	137	27.235457063711912	31.2	0
646	167	17.0	23.4	1
462	74	40.0	35.3	0
389	100	23.0	31.6	0
72	126	33.0	43.4	1
645	157	35.0	39.4	0

Correlation Heatmap - Pima Indians Diabetes Dataset



Split Train Test



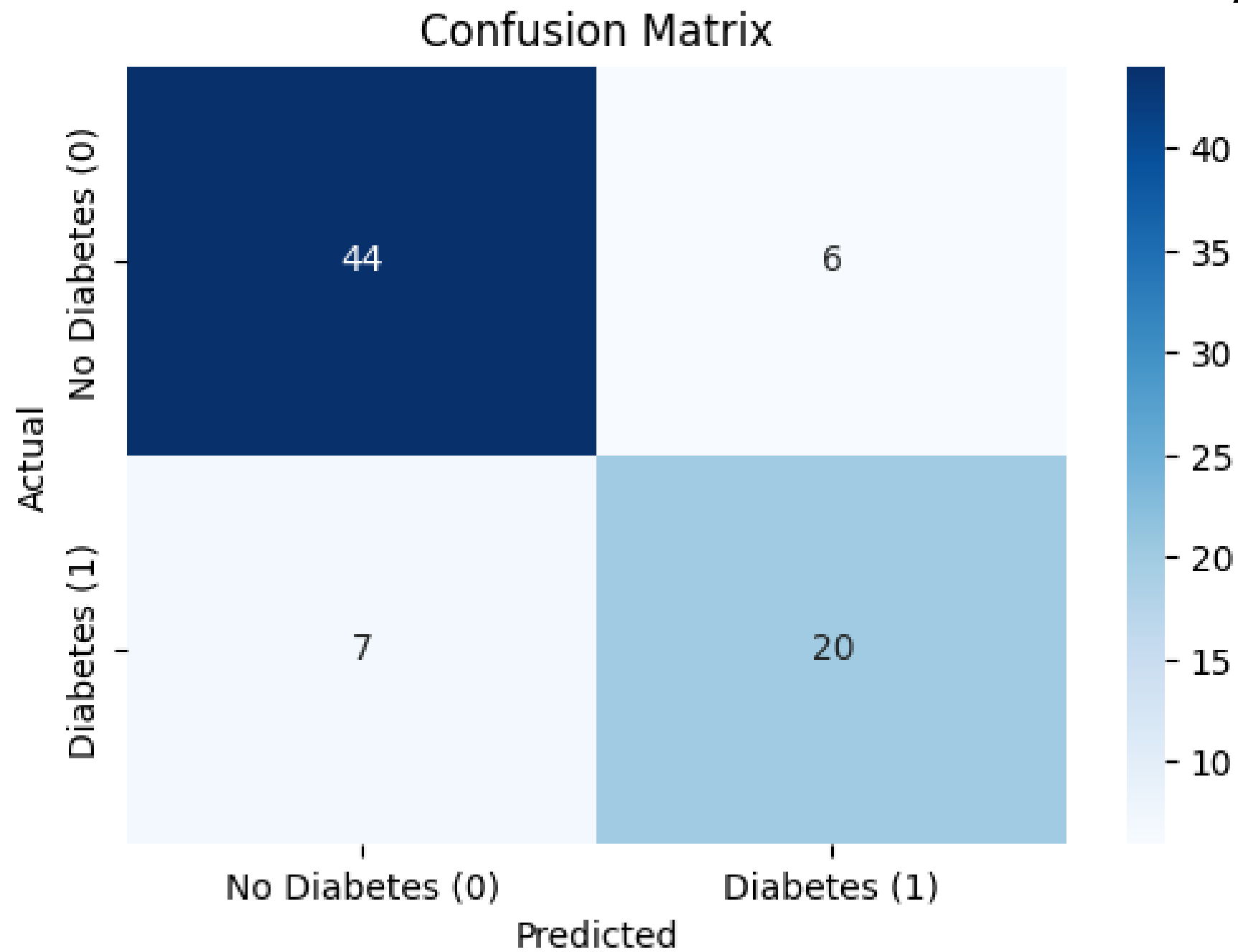
```
=== Dataset Split Summary ===
Total samples: 768
Training samples: 691 (90.0%)
Test samples: 77 (10.0%)

=== Outcome Distribution ===
Original dataset:
Outcome
0    0.651042
1    0.348958
Name: proportion, dtype: float64
  No Diabetes (0): 500 samples
  Diabetes (1): 268 samples

Training set:
Outcome
0    0.65123
1    0.34877
Name: proportion, dtype: float64
  No Diabetes (0): 450 samples
  Diabetes (1): 241 samples

Test set:
Outcome
0    0.649351
1    0.350649
Name: proportion, dtype: float64
  No Diabetes (0): 50 samples
  Diabetes (1): 27 samples
```

Finally SVM!



--- Classification Report ---

	precision	recall	f1-score	support
No Diabetes (0)	0.86	0.88	0.87	50
Diabetes (1)	0.77	0.74	0.75	27
accuracy			0.83	77
macro avg	0.82	0.81	0.81	77
weighted avg	0.83	0.83	0.83	77

=== Model Evaluation ===

Accuracy: 0.8312 (83.12%)

--- Confusion Matrix ---

```
[[44  6]
 [ 7 20]]
```

Conclusion



1. Based on this dataset the importance data is Glucose, Skin Thickness, BMI
2. Accuracy of SVM using this variable is 83.12%



"If god is statistician, then everything and everywhere you see is data."